Emergency Vehicle Sensing and Communication Network (ISAC-V2X)

An ISAC-V2X Framework for Traffic-Density Sensing and Emergency Vehicle Alerting

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Milestone 2: Problem Definition, Literature Review, SOTA Positioning, and Scope Finalization.

***Abstract*—**Emergency vehicles lose response time on congested routes, and nearby drivers may notice an approaching emergency vehicle (EV) only at short range. This project proposes an ISAC-based V2X framework in which a shared radio signal supports traffic-density estimation and carries emergency alerts. Roadside units estimate density per road segment by counting vehicle beacons, a route-optimization module based on Dijkstra or A* uses these counts to select a less-congested path, and the EV alerts vehicles on that path. Milestone 2 is a code-only simulation: beacon counting, density estimation, routing and alert delivery run in software on a virtual clock with seeded random loss, using PHY parameters from the OpenISAC design at **2.4 GHz**. The results test the software against the Milestone 1 criteria and are not radio measurements. USRP B210 validation, real density validation and on-air latency are planned for Milestone 3.

For the second milestone of the project, a simulation-based approach will be utilized that utilizes OpenISAC as our continuous-wave OFDM physical layer for the communication between the emergency vehicle and the roadside units. While the overall system architecture maintains that the emergency vehicle initiates the communication and the roadside units respond with the route optimization feedback prior to the emergency vehicle broadcasting the alert, the second milestone will instead focus upon simulating the communication between the emergency vehicle and the roadside units in code. This simulation will demonstrate the concept of the proposed system and make the code visible to the project team, temporarily bypassing the physical measurement of the traffic density on the road. Additionally, the simulation will establish the performance of the proposed physical layer. Hardware OTA validation and the implementation of the physical measurement of traffic density will be deferred to the third milestone of the project.

***Index Terms*—**Integrated sensing and communications, ISAC, V2X, emergency vehicle, DSRC, software-defined radio, GNU Radio, USRP B210, OFDM, traffic density, route optimization.

I. Introduction

Emergency response systems are dependent upon an emergency vehicle (EV) being able to move through the traffic without unnecessary delay. In many cases, the driver of the EV may not have sufficient information regarding the density of each of the road segments in the route that the EV needs to take to reach the emergency as quickly as possible. Furthermore, the vehicles in the traffic may not receive an early warning that an EV is approaching their road segment.

There are a variety of emergency-vehicle warning systems that have been developed and are in use today. For example, HAAS Alert provides responders-to-vehicles (R2V) and responders-to-responders (R2R) alerts that are based upon real-time vehicle information and cloud-based processing of that information. As such, the system is dependent upon the vehicles being connected, the use of navigation applications, GPS, cellular communications, and cloud processing of the information.

DSRC has been demonstrated for vehicle-to-vehicle safety warnings [2], and C-V2X is reported to outperform DSRC/802.11p beyond about 100 m [3]. ISAC-enabled vehicular networks use the same radio resources for sensing and communication [4]. OpenISAC [5], [6] is an open-source real-time OFDM-ISAC platform for USRPs, and this project takes its frame design as the basis for a software simulation. We did not run the OpenISAC stack in Milestone 2. Configuration overlays for its channel simulator were prepared but have not been checked against an OpenISAC build.

II. Problem Definition

The project problem is defined by three related operational challenges:

A. Emergency-vehicle response latency

An EV may lose critical response time when it enters a congested road corridor. Without a live estimate of the density of alternative road segments, the driver or routing system cannot reliably determine whether the current route remains preferable.

B. Route congestion blindness

The EV requires a mechanism that converts radio observations into a useful estimate of traffic density. The proposed project does not attempt to reconstruct the complete physical traffic state. Instead, it uses the number of active beacon signals observed on a road segment as a low-cost density proxy. This proxy is then used to update the cost of the corresponding road segment in a routing graph.

C. Belated civilian awareness

Civilian drivers travelling along the EV's selected route may become aware of the emergency vehicle only when visual or audible warning cues become noticeable. The proposed system therefore includes a direct RF broadcast mechanism through which the EV can send a targeted "make way" alert to nearby nodes.

D. Core problem statement

The project seeks to design and validate a lightweight two-way RF system in which a shared physical link supports both traffic-density sensing and V2X communication. The intended closed loop is:

$$\begin{aligned} \text{EV intent} &\rightarrow \text{RF density sensing} \\ &\rightarrow \text{route selection} \\ &\rightarrow \text{RF alert broadcast.} \end{aligned} \tag{1}$$

The shared-channel system is the centerpiece of our project.

III. Proposed System

A. System concept

The architecture has three modules: density sensing, route optimization and alert delivery. RSUs and relay nodes count vehicle beacons per road segment as a coarse density estimate. The router converts the counts into edge costs and computes the EV's path with Dijkstra or A*. The alert module then sends a "make way" alert to nodes on the selected route. In Milestone 2 this loop runs in software, with a virtual clock for beacons and in-process loopback links for alerts.

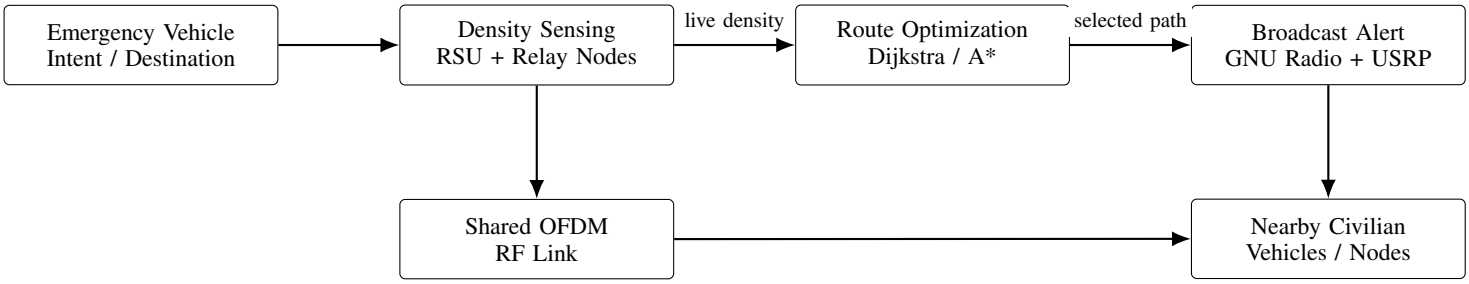


Fig. 1. Proposed ISAC-V2X system architecture showing the density sensing, route optimization, and broadcast alert modules.

B. Density sensing module

The density module is based on beacon counting. Each road segment is associated with one or more sensing/relay nodes. The number of active beacon signals observed during a measurement interval is treated as a proxy for the number of vehicles present on that segment. This approach is deliberately lightweight: the project does not require a dedicated radar sensor or detailed object classification at the Milestone 2 stage.

The estimate is compared with the simulator's known true count. Validation against manually counted traffic needs physical tests and is deferred to Milestone 3. Beacon counting is cooperative. Only vehicles with a beacon are counted, so it under-counts in mixed traffic, a silent or spoofed transmitter cannot be detected, and speed is available only from reported values. In the lab simulation every test vehicle beacons, so a good result says nothing about real traffic (running the demo with `--equipped-fraction 0.5` shows the failure). An optional radar-based counter, which counts moving targets in OpenISAC range-Doppler maps, is implemented but was tested only on synthetic frames, because the sensing-message decoder has not been written. It merges vehicles within one range bin (7.5 m at 20 MHz) and reads a stationary queue as empty.

C. Route optimization module

The routing module converts the measured density of each road segment into a dynamic edge cost. The project plan identifies Dijkstra and A* as the candidate routing algorithms. The selected algorithm will recompute the emergency vehicle path when live density information changes.

The routing layer is intentionally separated from the RF physical layer so that it can first be tested using synthetic or recorded density values. This modular structure reduces the difficulty of debugging the complete system and permits independent verification of the routing logic.

D. Alert and Handshake module

Once a route is selected, the EV sends a "make way" alert to nodes on that route through a hub and per-segment relays. Receivers identify EV alerts by a frame header protected by a CRC, and frames that fail the check are rejected. In the demo the hub forwards every alert to every receiver, and per-segment relay forwarding is covered by unit tests. The three receiving nodes are virtual relays, since OpenISAC is a point-to-point BS ¡-¿ UE link. The 90% PDR target is checked in simulation only. An ACK frame back to the EV and a request/acknowledge exchange with retries are not implemented in Milestone 2 and are future work.

E. Physical-layer concept

The physical layer uses the OpenISAC continuous-wave OFDM architecture. The frame structure consists of one full-band Zadoff-Chu synchronization symbol per frame, pilot subcarriers in every data symbol, and QPSK data mapped with LDPC coding.

The receiver operates in two states: `SYNC_SEARCH` and `NORMAL`, with synchronization detected via a normalized correlation and a peak-to-average threshold. While inspired by NR, OpenISAC is not strictly standard-compliant. The simulation is configured for 2.4 GHz, matching one of OpenISAC's standard frequency options. Deployments at 5.8 or 5.9 GHz are reserved as future hardware options.

TABLE I
MILESTONE 2 PHY SIMULATION PARAMETERS

Parameter	**Value**
Carrier	2.4 GHz
Bandwidth / sample rate	20 MHz
Subcarriers N	1024 (spacing 19.53kHz)
Cyclic prefix	128 samples (6.4 μs)
Symbols per frame M	100 (frame $\sim$5.76 ms)
Modulation / coding	QPSK, LDPC
Channel model	Noise, delay, frequency and timing offset; Doppler set to 0

IV. State of the Art and Related Work

Milestone 2 uses three representative benchmarks to position the proposed system: HAAS Alert, DSRC-based rear-end collision warning, and ISAC-enabled vehicular networks. Capability: open-source real-time OFDM-ISAC platform on USRPs (continuous-wave OFDM, monostatic and bistatic sensing, optional uplink per its YAML reference). Limitation: a general platform with no vehicular application layer (routing, EV alerting). .The comparison is summarized in Table II.

A. HAAS Alert

The HAAS Alert report describes two main use cases: responder-to-vehicle (R2V) alerts for civilian drivers and

TABLE II
SUMMARY OF RELATED WORK AND PROJECT POSITIONING.

Benchmark	Main capability	Strength	Limitation relative to this project
HAAS Alert [1]	R2V and R2R emergency alerts using real-time vehicle data, cellular connectivity, and cloud processing.	Demonstrates operational EV warning concepts and evaluates usability.	Does not combine RF traffic-density sensing with the alerting loop; depends heavily on cloud/cellular connectivity.
OpenISAC *et al.* [2]	Real-time continuous-wave OFDM-ISAC platform evaluated on USRPs.	Provides a robust, open-source foundation for shared waveform processing.	Downlink-focused; lacks a dedicated vehicular request/acknowledge handshake layer.
Proposed System	OpenISAC PHY model + dynamic routing + REQUEST/ACK/FEEDBACK handshake over shared OFDM link.	Combines sensing proxies, routing, and robust end-to-end V2X handshake into one closed loop.	Simulation-only for Milestone 2; hardware sensing deferred to M3.

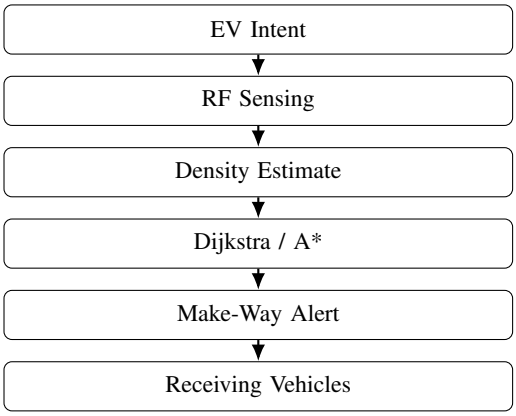


Fig. 2. Closed-loop operating sequence of the proposed ISAC-V2X system.

responder-to-responder (R2R) alerts between equipped emergency vehicles [1]. In the R2V case, alerts are associated with the predicted path of an emergency vehicle and can be delivered through navigation applications or vehicle systems. In the R2R case, the system identifies potential collisions between emergency vehicles connected to the HAAS Safety Cloud.

The system therefore establishes an important benchmark for the alerting component of the proposed project. Its evaluation also demonstrates that operational suitability includes more than communication accuracy: alert timing, usability, configuration, reporting, and operator interaction are relevant considerations.

B. OpenISAC

OpenISAC provides an open-source, real-time OFDM-ISAC processing framework. While existing implementations demonstrate successful synchronization and continuous-wave OFDM processing on USRPs, they are primarily focused on standard physical layer ISAC performance metrics rather than targeted V2X application layers. OpenISAC [5] uses continuous-wave OFDM with a Zadoff-Chu sync symbol, pilots and LDPC-coded QPSK. Its authors report error-free reception down to about 7.8 dB SNR on X310-to-B210 hardware at 3.1 GHz and 50 MHz [5, Fig. 11]. The code is released under BSD-2-Clause [6].

V. RESEARCH GAP AND PROJECT NOVELTY

Open OFDM-ISAC platforms such as OpenISAC [5] provide the physical layer and sensing pipeline, but none of the work we reviewed combines them with a vehicular application layer that turns shared-link density estimates into emergency-vehicle routing and alerting. Our contribution is that layer: beacon-based density estimation, Dijkstra/A* routing, EV alert delivery and a software evaluation harness. We do not claim a new PHY.

Code Attribution: The simulation environment uses a Python-based broadcasting, routing and sensing framework built on OpenISAC.

VI. SCOPE, ASSUMPTIONS, AND CONSTRAINTS

A. Scope

The scope finalized for the project is:

- Implement the OpenISAC model in over-the-air (OTA) simulation
- Isolate and simulate the REQUEST/ACK/FEEDBACK handshake between the OBU and RSU
- Execute dynamic route selection (Dijkstra/A*) using synthetic denisty inputs.
- Measure simulated PDR, ACK success, and retry statistics.
- Implement Dijkstra/A* based dynamic route selection;
- Transmit emergency “make way” alerts to vehicles/nodes on the selected route;
- Support at least three receiving nodes in the RF demonstration;
- Identify EV alert signals using a preamble/header and evaluate false-positive and false-negative behaviour;
- Integrate the modules into an end-to-end laboratory demonstration.

B. Assumptions

For Milestone 2, the following assumptions are made:

- Vehicles are quasi-static within a single frame duration.
- Doppler shift is neglected (assumed to be 0Hz) as it is small relative to the subcarrier spacing.
- The wireless environment is represented via a synthesized channel model (noise, delay, frequency/timing offsets).

TABLE III
PROJECT EVALUATION CRITERIA.

Criterion	Target / assessment method
RF reliability	Demonstrate at least 90% PDR across receiving nodes.
Density proxy validity	Compare beacon-count estimates with manual ground truth in laboratory scenarios.
Route adaptation	Verify that the Dijkstra/A* recomputes path when synthetic density changes.
End-to-end operation	Demonstrate EV intent → synthetic sensing → routing feedback → alert broadcast.
Signal identification	Measure false-positive and false-negative behaviour of the sync threshold..
Latency	Compare alert latency and time saved.

- Traffic density inputs are synthetically generated for the M2 routing logic.
- All test vehicle carry beacon
- Density Inputs are simulated
- Virtual relay nodes can be used when physical nodes are insufficient for early testing;
- The project's functionality is evaluated as a prototype rather than a production-certified emergency response system.

C. Constraints

We haven't validated any real hardware (USRP) in this Milestone 2. Also channel conditions are simplified (0 Hz Doppler). Currently our simulation is also bound by OpenISAC's host-processing limits. No USRP hardware was used in testing so far, though the code should run over the air. Latency is software time only on loopback, and says nothing about OFDM framing or USRP buffering. Thresholds in `eval.*` are placeholders chosen in code. OpenISAC's datagram framing, control protocol and multi-UE support are undocumented, so unverified. We currently use overlay configs for the USRPs. As of now these have not been checked against an OpenISAC build, which will be done in the hardware verification as part of the next milestone.

VII. CRITERIA FOR ASSESSING THE PROPOSED SYSTEM

The project requires measurable criteria so that later milestones can determine whether the proposed architecture is viable. Table III defines the principal criteria.

Our core objectives include PDR, density-proxy validation, dynamic routing, end-to-end integration, and signal identification. Performance comparisons of sensing-enabled and non-sensing cases are also included among the expected deliverables.

VIII. RESEARCH METHODOLOGY

Milestone 2 is primarily a design and research milestone. Our methodology therefore focused on establishing a traceable link between the identified problem, the literature, the proposed architecture, and the measurable criteria.

A. Literature review

The literature review uses four benchmark categories. The HAAS Alert report was used to understand an operational emergency-vehicle warning architecture and its evaluation dimensions. The DSRC study was used to understand direct RF vehicle-safety communication and the role of transmission delay and positioning errors. The ISAC review was used to establish the architectural principle of jointly using radio resources for sensing and communication. The C-V2X and DSRC performance analysis study was used to make decisions about the standard used for transmission and the performance limitations our system will have to work around.

B. System decomposition

The proposed system is decomposed into independent modules:

1) density sensing,
2) route optimization,
3) RF broadcast alerting,
4) signal identification, and
5) end-to-end integration.

This decomposition permits each software or DSP block to be tested independently before the final integration stage.

C. Density validation methodology

Because physical density validation via manual vehicle counting cannot be executed in code, the Milestone 2 simulation relies on synthetic density data to validate the algorithm's decision-making and route-switching logic. Real-world validation against physical manual ground truth is deferred to Milestone 3.

D. Simulation methodology

The Milestone 2 evaluation is entirely software-driven. PDR is measured by simulating multiple packet transmissions across varying SNR points to generate BER/BLER curves, which are evaluated against standard OpenISAC performance metrics. The REQUEST/ACK handshake is evaluated by tracking ACK success rates and analyzing retry statistics under different simulated channel impairments. Furthermore, we conduct a synchronization threshold sweep to measure false-positive and false-negative detection rates during the `SYNC_SEARCH` state.

E. Route and alert validation

The route module will first be tested using generated density maps. Once the route logic is stable, it will receive live or recorded density updates. The selected route will then determine the intended alert recipients. The RF layer will be evaluated using packet delivery ratio, signal identification accuracy, and alert latency.

TABLE IV
MILESTONE ROADMAP.

Phase	Focus	Expected output
Milestone 1	Problem and SOTA positioning	Problem definition, literature review, SOTA comparison, research gap, scope, assumptions, evaluation criteria, and finalized architecture.
Milestone 2	Code-only Simulation	OpenISAC PHY simulation, handshake logic, synthetic routing tests.
Milestone 3	System Integration	Hardware validation (USRP B210), real-world density validation, latency and time-saved metrics.
Final phase	Evaluation and report	Final documented PHY parameters, signal-ID results, logs, and defense.

IX. Milestone Plan and Expected Deliverables

Milestone 2 (current) delivers a code-only simulation: density estimation, routing, alert delivery, signal-identification tests and the evaluation script. Milestone 3 adds USRP B210 hardware validation, a 2.4 GHz spectrum survey, density validation against manual counts, on-air latency, the radar estimator's sensing-message decoder, and, if time allows, the ACK and feedback path.

X. Conclusions and Future Outlook

Milestone 2 creates a tested software base for an ISAC-V2X emergency vehicle sensing and communication prototype. The project looks at three problems: delays in emergency response no live information about how busy roads are and slow alerts to people who are not involved. The solution is a loop that works both ways of sending messages one way. It uses a two-way REQUEST/ACK/FEEDBACK V2X process along with changing how routes are chosen. The review of what's already available shows that current systems have useful parts but there is a need for complete solutions that work at the edge with radio frequency. HAAS Alert shows how to warn emergency vehicles from the cloud and let responders send alerts to each other. OpenISAC gives a way to process data in real time using continuous-wave OFDM-ISAC. The project brings these together by building a part for vehicles on top of a physical layer inspired by OpenISAC. This allows route feedback over a shared radio signal. By using a software-simulation at 2.4 GHz for this step the project made sure the main communication and deciding part worked well. Testing with data made it easy to check the Dijkstra and A^* routing methods as well as how well the OBU and RSU parts work together without dealing with real hardware problems. This way of testing gave numbers like how many messages get through how often they are accepted and how well the system knows when things are in sync. The main plan for Milestone 3 is to keep the separate parts as the project moves from software to real equipment. Since the way routes are chosen and the radio part have been tested in code any problems during setup can be found and fixed step by step. At the end of Milestone 2 the project has a tested part, for vehicles, a simulated radio part and a clear way to use USRP radios to check how the whole system works in real life. This includes checking how well it deals with moving signals how accurately it measures roads and how fast it works.

References

[1] M. Monetti, R. Casparro, K. Dooley, and T. Mackanin, *Emergency Vehicle Warning System (HAAS Alert): Virtual Technology Demonstration Report*, National Urban Security Technology Laboratory, U.S. Department of Homeland Security, Science and Technology Directorate, July 2021.

[2] X. Zhao, S. Jing, F. Hui, R. Liu, and A. J. Khattak, "DSRC-based rear-end collision warning system – An error-component safety distance model and field test," *Transportation Research Part C: Emerging Technologies*, vol. 107, pp. 92–104, 2019, doi: 10.1016/j.trc.2019.08.002.

[3] H. I. Abbasi, R. Gholmieh, T. V. Nguyen, S. Patil and J. Misener, "LTE-V2X (C-V2X) Performance in Congested Highway Scenarios," *ICC 2022 - IEEE International Conference on Communications*, Seoul, Korea, Republic of, 2022, pp. 303-308, doi: 10.1109/ICC45855.2022.9838706.

[4] Z. Du et al., "Toward ISAC-Empowered Vehicular Networks: Framework, Advances, and Opportunities," in *IEEE Wireless Communications*, vol. 32, no. 2, pp. 222-229, April 2025, doi: 10.1109/MWC.002.2300496.

[5] Z. Zhou, C. Zhang, X. Xu, and Y. Zeng, "OpenISAC: An open-source real-time experimentation platform for OFDM-ISAC," arXiv:2601.03535v2 [eess.SP], Jul. 2026. [Online]. Available: https://arxiv.org/abs/2601.03535

[6] Z. Zhou, "OpenISAC," GitHub repository (BSD-2-Clause licence). [Online]. Available: https://github.com/zhouzhiwen2000/OpenISAC.

[7] Government of India, Ministry of Communications, Department of Telecommunications (Wireless Planning and Coordination Wing), *(title of the notification)*, *(G.S.R. number)*, *(date)* — de-licensing of the 2.4 GHz band.